

Class 02

Database & Table Creation

In this session, we'll learn about the basic CRUD operations in postgresSQL and then move a little beyond them.

- **CREATE DATABASE**
- **USE DATABASE**
- **CREATE TABLE**
- **Data Types**
- **Viewing structure**

CREATE DATABASE

So, very well done till here we have understood what is DBMS, what is RDBMS now let's start working with postgresQL.

Before we start writing queries, we first need to create a database where all our tables and data will be stored.



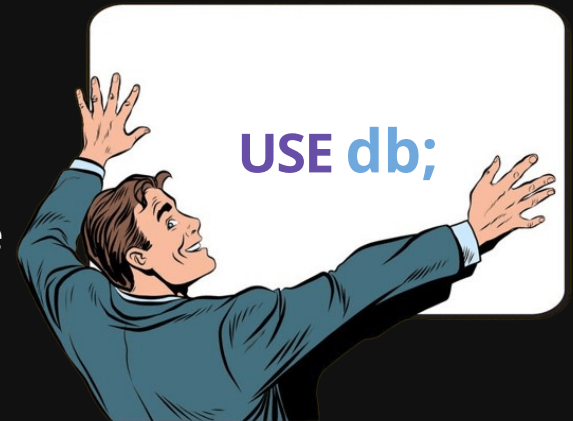
- Here CREATE and DATABASE are keywords of postgresQL
- And "db" is the name of database.

USE DATABASE

As we have created a database where now we can finally create table and insert our data.

But, before creating tables and inserting data, we first need to select which database we want to work with.

We use "USE" to select the database where we want to create our table and where we want to store our data.



CREATE TABLE

Now that we have selected our database, the next step is to create a table.

Tables are the backbone of a database – they store data in the form of rows (records) and columns (fields).



```
CREATE TABLE table_name (  
    column1 datatype constraint,  
    column2 datatype constraint,  
    ... );
```

Data Types

DATE TYPE	SPEC
CHAR	String (0 - 255)
VARCHAR	String (0 - 255)
TINYTEXT	String (0 - 255)
TEXT	String (0 - 65535)
BLOB	String (0 - 65535)
MEDIUMTEXT	String (0 - 16777215)
MEDIUMBLOB	String (0 - 16777215)
LONGTEXT	String (0 - 4294967295)
LOBLOB	String (0 - 4294967295)
TINYINT	Integer (-128 to 127)
SMALLINT	Integer (-32768 to 32767)
MEDIUMINT	Integer (-8388608 to 8388607)

DATA TYPE	SPEC
INT	Integer (-2147483648 to 2147483647)
BIGINT	Integer (-9223372036854775808 to 9223372036854775807)
FLOAT	Decimal (precise to 23 digits)
DOUBLE	Decimal (24 to 53 digits)
DECIMAL	"DOUBLE" stored as string
DATE	YYYY-MM-DD
DATETIME	YYYY-MM-DD HH:MM:SS
TIMESTAMP	YYYYMMDDHHMMSS
TIME	HH:MM:SS
ENUM	One of preset options
SET	Selection of preset options
BOOLEAN	TINYINT(1)

Viewing Structure

Once a table is created, we may want to see its structure (like columns, data types, and constraints). For this, postgresQL provides two commands.

1. **DESCRIBE** command



DESCRIBE table name;

Shows column names, data types, whether `NULL` is allowed, keys, default values, etc.

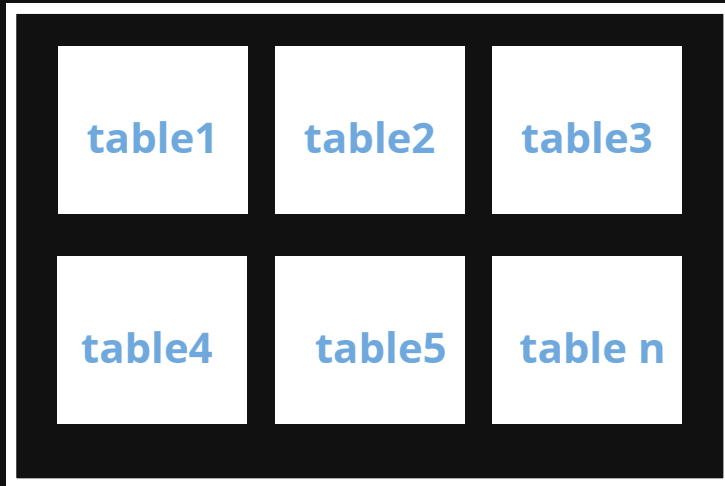
2. **SHOW COLUMNS** command



SHOW COLUMNS FROM table name;

Shows column names, data types, whether `NULL` is allowed, keys, default values, etc.

db;



This we already know that a database consist of multiple tables

This is how a table is created inside our database, and each column in the table is defined with its own data type.

table 1

col1	col2	col3	col4
-	-	-	-
-	-	-	-
-	-	-	-

A sample table from our database.